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SECURITY PROJECT

Miller-Rabin, the Theory

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Contents

1	Introduction	2
2	Miller-Rabin primality test	3
2.1	Fermat's Test	3
2.2	The Proportion of Miller-Rabin witnesses	6

1 Introduction

In this paper we explore an important theorem concerning the Miller–Rabin primality test and study its proof. As we shall see, the theorem states that the probability of a composite number n passing the Miller-Rabin test is very low. Therefore, in practice, n successfully passing a handful of Miller-Rabin tests quickly convinces one of its primality. This is sufficient for most commercial applications.

For better understanding, we first begin by introducing a similar but much simpler result concerning the Fermat primality test. By Fermat’s test, if n is prime, then for all positive integers x we have $x^{n-1} = 1 \pmod{n}$. This suggests the Fermat test for a prime is as follows: pick a random $x \in \{1, \dots, n-1\}$ and check whether $x^{n-1} = 1 \pmod{n}$. If not, then n must be composite.

Recalling that n is prime if and only if the solutions of $x^2 = 1 \pmod{n}$ are $x = \pm 1$ we can instead apply the Miller-Rabin primality test. In precise sense, the Miller-Rabin test is not a primality test, but rather a ‘compositeness test’, since it proves accurately if a given number is composite and, probabilistically, suggests primality.

For an odd composite number $n > 1$, less than 25% of numbers from 1 to $n-1$ are nonwitnesses in the Miller-Rabin test. In the upcoming sections, we are going to prove the 25% upper bound, in order to illustrate the likeliness of a randomly chosen number n to be prime.

2 Miller-Rabin primality test

Suppose we want to test whether an integer $n > 1$ is prime or not. The Miller-Rabin test goes as follows:

Write $n-1 = 2^k m$, where $m, k \in \mathbb{Z}_{\geq 1}$ and m is odd. Choose a random $0 \neq x \in \mathbb{Z}/n\mathbb{Z}$ and compute $b = a^m \pmod{n}$. If $b \equiv 1 \pmod{n}$, choose another x , otherwise compute the sequence $b, b^2, \dots, b^{2^{k-1}}$. If it contains $-1 \pmod{n}$, choose another x and repeat the previous steps. If the sequence does not contain $-1 \pmod{n}$, then n is composite and the test is terminated.

If the algorithm yields that n is composite, then it surely is. However, if the algorithm outputs that n is probably prime, it means that every x considered satisfies either $x^m = 1 \pmod{n}$, or $x^{m2^i} = -1 \pmod{n}$, where $0 \leq i \leq k$.

Every x that proves compositeness of n is relatively prime to n and is known as a Miller-Rabin witness. Additionally, it will become evident that the Fermat primality test possesses witnesses of its own. As a matter of fact, over 50% of the integers from 1 to $n-1$ are Fermat witnesses (2.1) and over 75% are Miller-Rabin witnesses, which is presented in 2.2.

2.1 Fermat's Test

We first introduce the simpler primality test, namely the Fermat test. Fermat's test makes a heavy use of Fermat's Little theorem, which involves congruent modulo prime numbers, by translating it to composite numbers.

Theorem 2.1.1 (Fermat's little theorem). *If p is prime, then $a^p \equiv a \pmod{p}$ for every $a \in \mathbb{Z}$.*

An integer less than n , which proves n is composite, is called a witness to the compositeness of n . The type of witness depends on the type of test we use.

Definition 2.1.2. Let $n \geq 2$ and $1 \leq b \leq n-1$ be integers. We call b a gcd witness for n if $\gcd(b, n) > 1$. We call b a Fermat witness for n if $b^{n-1} \not\equiv 1 \pmod{n}$.

Moreover, one can show that the two are related to each other. In fact, any gcd witness is a Fermat witness since: $\gcd(b, n) > 1 \Rightarrow b^{n-1} \not\equiv 1 \pmod{n}$.

What makes the Fermat witnesses so important is the fact that there are usually many more of them than gcd witnesses. The next theorem gives a bound on the number of Fermat witnesses for a number $n \geq 2$:

Theorem 2.1.3. Let $n \geq 2$. If some $b \in \mathbb{Z}$ satisfies $b^{n-1} \not\equiv 1 \pmod{n}$ and $\gcd(b, n) = 1$, then $|\{1 \leq a \leq n-1 : a^{n-1} \not\equiv 1 \pmod{n}\}| > \frac{n-1}{2}$.

That is, if some Fermat witness for n is relatively prime to n , then over half integers from 1 to $n-1$ are Fermat witnesses for n .

Proof. Consider the sets

$$A := \{1 \leq a \leq n-1 : a^{n-1} \equiv 1 \pmod{n}\},$$

$$B := \{1 \leq a \leq n-1 : \gcd(a, n) = 1 \text{ and } a^{n-1} \not\equiv 1 \pmod{n}\},$$

$$C := \{1 \leq a \leq n-1 : \gcd(a, n) > 1\}.$$

Let $a \in A$, then $a^{n-1} \equiv 1 \pmod{n}$, so a cannot be in B . The fact that $a^{n-1} \equiv 1 \pmod{n}$ implies $\gcd(a, n) = 1$, hence a is not in C . Furthermore, B and C are clearly disjoint.

So, the sets A , B and C are pairwise disjoint, and their union represents all integers from 1 to $n-1$.

Recall that any gcd witness is a Fermat witness. Notice that the set C is the set of gcd witnesses and B is the rest of Fermat witnesses. Hence, $B \cup C$ is the set of all Fermat's witnesses for n . A is everything else between 1 and $n - 1$:

$$1 \equiv 1 \pmod{n} \Rightarrow 1 \in A, 1 \notin B \cup C.$$

Note that here " $\cup$ " denotes the disjoint union, i.e. $B \cap C = \emptyset$.

By assumption, B is nonempty, so n must be composite by Fermat's little theorem.

This implies that C is nonempty as well since there exists some $a \in \{1, \dots, n - 1\}$ such that $\gcd(a, n) > 1$.

Claim: If $B \neq \emptyset$, then $|B| + |C| > \frac{n-1}{2}$.

Proof of the claim. To prove this, multiply each number in some set by a common constant.

Let $b \in B$ and consider

$$Ab := \{ab \pmod{n} : a \in A\}$$

Then $Ab \subset B$, since $(ab)^{n-1} = a^{n-1}b^{n-1} = b^{n-1} \not\equiv 1 \pmod{n}$

so $ab \pmod{n} \in B$.

Now suppose that $a, a' \in A$.

If $ab \equiv a'b \pmod{n}$ then by cancelling b , one obtains $a \equiv a' \pmod{n}$, so $a = a'$, as

$$1 \leq a, a' \leq n - 1.$$

This shows that $|Ab|$ and $|A|$ have the same cardinality and since $Ab \subset B$, we obtain

$|A| = |Ab| \leq |B|$. Then,

$$n - 1 = |A \cup B \cup C| = |A| + |B| + |C| \geq |A| + |A| + 1 > 2|A|$$

$$\Rightarrow |A| < \frac{n-1}{2}$$

which means A is less than half of $\{1, \dots, n-1\}$, so its complement $B \cup C$ is more than half of $\{1, \dots, n-1\}$:

$$|B \cup C| = |B| + |C| = n-1 - |A| > n-1 - \frac{n-1}{2} = \frac{n-1}{2}$$

□

2.2 The Proportion of Miller-Rabin witnesses

When presented with a composite integer $n > 1$, the algorithm rapidly proves its nonprimality. Conversely, if n “passes” the test, it is most likely prime. The algorithm iteratively executes the Miller-Rabin test numerous times, employing distinct random initializations on each iteration.

Recall the strategy introduced for the Fermat test:

if $a^{n-1} \not\equiv 1 \pmod{n}$ for at least one integer x , $x \in \{1, \dots, n-1\}$, then n is composite.

Indeed, if n were prime then Fermat’s little theorem says $x^{n-1} \equiv 1 \pmod{n}$ for *all* integers $x \in \{1, \dots, n-1\}$. When this breaks down even once, n must be composite, and the integer less than n that proves n ’s compositeness is a Fermat witness.

We introduce Miller-Rabin witnesses. Similarly to Fermat, the term “witness” refers to a number that proves n is composite. If a Miller-Rabin witness is found, then n must be composite since an odd prime cannot have a Miller-Rabin witness.

Definition 2.2.1. For positive odd integer $n > 1$, write $n-1 = 2^k m$ with m odd and pick $x \in \{1, \dots, n-1\}$. We say x is a Miller-Rabin witness for n if the following

holds:

$$x^m \not\equiv 1 \pmod{n} \text{ and } x^{m2^i} \not\equiv -1 \pmod{n} \text{ for all } i \in \{0, \dots, k-1\}.$$

We say x is a Miller-Rabin nonwitness for n if one of the following congruences holds:

$$x^m \equiv 1 \pmod{n} \text{ or } x^{m2^i} \equiv -1 \pmod{n} \text{ for all } i \in \{0, \dots, k-1\}.$$

In order to introduce the main theorem, we need one more ingredient:

Theorem 2.2.2 (Chinese remainder theorem). *For any integer $n \geq 1$, if $n = n_1 \dots n_r$ where the n_i 's are pairwise coprime, $i \in \{1, \dots, r\}$, that is $\gcd(n_i, n_j) = 1$, for all $i \neq j$, then we have a group isomorphism*

$$\mathbb{Z}/n\mathbb{Z} \cong \mathbb{Z}/n_1\mathbb{Z} \times \dots \times \mathbb{Z}/n_r\mathbb{Z}.$$

Moreover, the same map induces a group isomorphism between $(\mathbb{Z}/n\mathbb{Z})^\times$ and $(\mathbb{Z}/n_1\mathbb{Z})^\times \times \dots \times (\mathbb{Z}/n_r\mathbb{Z})^\times$.

Theorem 2.2.3. *Let $n > 9$ be an odd positive composite integer. We write $n-1 = 2^k m$ for some exponent $k \geq 1$ and some odd integer m . Let*

$$B := \{x \in (\mathbb{Z}/n\mathbb{Z})^\times : x^m = 1 \text{ or } x^{m2^i} = -1 \text{ for some } 0 \leq i \leq k\}.$$

Then we have

$$\frac{|B|}{\varphi(n)} \leq \frac{1}{4}.$$

Note that here B represents the set of all nonwitnesses and $\varphi(n) = |(\mathbb{Z}/n\mathbb{Z})^\times|$ denotes Euler's totient function.

Proof. Let $n > 9$ be an odd composite integer such that $n - 1 = 2^k m$, for some exponent $k \geq 1$ and some positive odd integer m . Note that n can be decomposed as a product of t primes, including multiplicity, where $t \in \mathbb{N}$. So $n = \prod_{j=1}^t p_j^{a_{p_j}}$ where each p_j is prime and a_{p_j} is the exact power of p_j that divides n .

Consider B as defined in the hypothesis. Let 2^l denote the largest power of 2 with the property that it divides $p - 1$ for every prime divisor p of n and define:

$$B' := \{x \in (\mathbb{Z}/n\mathbb{Z})^\times : x^{m2^{l-1}} = \pm 1\}$$

Claim: B is contained in B' .

Proof of the claim. Let $x \in B$. Then $x^m = 1$ or $x^{m2^i} = -1$, for $0 \leq i \leq k$. If $x^m = 1$, then $x^{m2^{l-1}} = 1^{2^{l-1}} = 1$ and hence $x \in B'$.

Suppose now that $x^{m2^i} = -1$, then n divides $x^{m2^i} + 1$. Since every p divides n , p must divide $x^{m2^i} + 1$, that is, $x^{m2^i} \equiv -1 \pmod{p}$. Squaring both sides of the equality we obtain $x^{m2^{i+1}} \equiv 1 \pmod{p}$ and thus $o \mid m2^{i+1}$, where $o := \text{ord}(x)$. By Fermat's Little Theorem, the order of x divides $p - 1 = \varphi(p)$. Now, assume that $o \nmid m2^i$. Then $m2^i = c \cdot o$ for some $c \in \mathbb{Z}$. So $x^{m2^i} = x^{c \cdot o} \equiv 1 \pmod{p}$ which contradicts $x^{m2^i} \equiv -1 \pmod{p}$. Hence the order of x does not divide $m2^i$. It follows that the order of x must be a multiple of 2^{i+1} , in other words $o = 2^{i+1} \tilde{m}$, for some $\tilde{m} \in \mathbb{Z}_{>0}$. Again, by Fermat's Little Theorem, $2^{i+1} \tilde{m} \mid p - 1$, so for every p the exact power of 2 dividing the order of $x \pmod{p}$ is equal to 2^{i+1} . Therefore, $l \geq i + 1$ for every prime divisor p of n . Since $x^{m2^i} \equiv -1 \pmod{p}$, if $l > i + 1$, then $x^{m2^{l-1}} \equiv 1 \pmod{p}$ and for $l = i + 1$, $x^{m2^{l-1}} \equiv -1 \pmod{p}$. Thus, x is an element of B' .

We conclude that $B \subseteq B'$. This proves the claim.

Consider the equation $x^{m2^{l-1}} = 1$, for $x \in (\mathbb{Z}/n\mathbb{Z})^\times$. By the Chinese Remainder Theorem, since $x = \prod_{j=1}^t p_j^{a_{p_j}}$, the product over p_j of the number of solutions to the equation $X^{m2^{l-1}} \equiv 1 \pmod{p_j^{a_{p_j}}}$ is equal to the cardinality of the solutions set of $x^{m2^{l-1}} = 1 \pmod{n}$. That is,

$$|\{x \in (\mathbb{Z}/n\mathbb{Z})^\times : x^{m2^{l-1}} = 1\}| = \prod_{j=1}^t |\{X \in (\mathbb{Z}/p_j^{a_{p_j}}\mathbb{Z})^\times : X^{m2^{l-1}} = 1\}|$$

Since $p_j > 2$, for all $j \in \{1, \dots, t\}$, each $(\mathbb{Z}/p_j^{a_{p_j}}\mathbb{Z})^\times$ is cyclic. Then there exists a primitive root g such that $g^{\varphi(p_j^{a_{p_j}})} = 1$. This means that every element in $(\mathbb{Z}/p_j^{a_{p_j}}\mathbb{Z})^\times$ can be uniquely represented as some power r of g , for $1 \leq r \leq \varphi(p_j^{a_{p_j}})$.

Then

$$x^{m2^{l-1}} = g^{rm2^{l-1}} \equiv 1 \pmod{p^{a_p}},$$

for all $p \in \{p_j : j = 1, \dots, t\}$ and hence $\varphi(p^{a_p})$ divides $rm2^{l-1}$.

Note that $\varphi(p^{a_p}) = (p-1)p^{a_p-1}$ and define $d := \gcd((p-1)p^{a_p-1}, m2^{l-1})$. Write $m2^{l-1} = m_1d$ and $\varphi(p^{a_p}) = n_1d$, then $\gcd(n_1, m_1) = 1$ and since $\varphi(p^{a_p}) | rm2^{l-1}$, we must have that n_1 divides rm_1 . Now, $\gcd(n_1, m_1) = 1$, so $n_1 | r$ and we find d solutions for r :

$$n_1, 2n_1, \dots, (d-1)n_1, dn_1 = \varphi(p^{a_p})$$

Therefore, the number of solutions to the equation $x^{m2^{l-1}} \equiv 1 \pmod{p^{a_p}}$ is given by $d = \gcd((p-1)p^{a_p-1}, m2^{l-1})$.

Since $n-1 = m2^k$ and $l \leq k$, $m2^{l-1}$ divides $n-1$, then $m2^{l-1}$ cannot divide p , so $m2^{l-1}$ does not divide p^{a_p-1} and, thus, $\gcd(p^{a_p-1}, m2^{l-1}) = 1$. This is equivalent

to

$$\gcd((p-1)p^{a_p-1}, m2^{l-1}) = \gcd(p-1, m2^{l-1}).$$

Notice that, by construction, $2^l | p-1$, so $2^{l-1} | p-1$, and m is odd, that is, we cannot have $2 | m$, then 2^{l-1} can be factored out from the previous equality:

$$\gcd(p-1, m2^{l-1}) = \gcd(p-1, m)2^{l-1}.$$

Therefore, each equation of the form $X^{m2^{l-1}} \equiv 1 \pmod{p^{a_p}}$ has precisely $\gcd(p-1, m)2^{l-1}$ solutions and

$$|\{x \in (\mathbb{Z}/n\mathbb{Z})^\times : x^{m2^{l-1}} = 1\}| = \prod_{p|n} \gcd(p-1, m)2^{l-1}.$$

Similarly, the number of solutions to $X^{m2^l} \equiv 1 \pmod{p^{a_p}}$ is equal to $\gcd(p-1, m)2^l$, which is twice the number of solutions to $X^{m2^{l-1}} \equiv 1 \pmod{p^{a_p}}$.

Note that

$$X^{m2^l} - 1 \equiv 0 \pmod{p^{a_p}} \iff (X^{m2^{l-1}} - 1)(X^{m2^{l-1}} + 1) \equiv 0 \pmod{p^{a_p}}.$$

Then, the number of solutions to $X^{m2^{l-1}} \equiv -1 \pmod{p^{a_p}}$ is given by

$$\gcd(p-1, m)2^l - \gcd(p-1, m)2^{l-1} = \gcd(p-1, m)2^{l-1}.$$

Thus, we have

$$|B'| = 2 \prod_{p|n} \gcd(p-1, m)2^{l-1}$$

and so

$$\frac{|B'|}{\varphi(n)} = 2 \prod_{p|n} \frac{\gcd(p-1, m)2^{l-1}}{(p-1)p^{a_p-1}}$$

since $\varphi(n) = \prod_{p|n} (p-1)p^{a_p-1}$

Aiming for contradiction, assume that $\frac{|B|}{\varphi(n)}$ is strictly greater than $\frac{1}{4}$. Noting that $B \subseteq B'$, i.e. $|B| \leq |B'|$, it is sufficient to show that

$$\frac{1}{4} < 2 \prod_{p|n} \frac{\gcd(p-1, m)2^{l-1}}{(p-1)p^{a_p-1}} \quad (1)$$

First note that $\frac{(p-1)/2}{\gcd(p-1, m)2^{l-1}} = \frac{p-1}{\gcd(p-1, m)2^l} \in \mathbb{Z}_{\geq 1}$, and, since $2^l | p-1$ and $\gcd(p-1, m) | p-1$, we have that $\gcd(p-1, m)2^{l-1}$ divides $\frac{p-1}{2}$. It follows immediately that $\frac{\gcd(p-1, m)2^{l-1}}{p-1} \leq \frac{1}{2}$ and, given that $p^{a_p-1} \geq 1$, it holds that $\frac{\gcd(p-1, m)2^{l-1}}{(p-1)p^{a_p-1}} \leq \frac{1}{2}$ for every prime divisor p of n . The right hand side of (1) is, therefore, at most $2 \cdot (\frac{1}{2})^t = 2^{1-t}$, where t is the number of different primes dividing n . It follows that $t \leq 2$.

Suppose that $t = 2$, so n has precisely two distinct prime divisors p and q . If one of them, say $p \geq 3$, has the property that p^2 divides n so that $a_p \geq 2$, then given that $\frac{\gcd(p-1, m)2^{l-1}}{(p-1)} \leq \frac{1}{2}$ and $\frac{1}{p^{a_p-1}} \leq \frac{1}{p}$ the right hand side of (1) is at most $\frac{2^{1-t}}{3} = \frac{1}{6}$. This contradicts the lower bound $\frac{1}{4}$.

It follows that all exponents a_p are equal to 1, so that $n = pq$ for two distinct primes p and q . The inequality (1) now becomes

$$\frac{1}{4} < 2 \prod_{p|n} \frac{\gcd(p-1, m)2^{l-1}}{(p-1)p^{a_p-1}} = \frac{2 \gcd(p-1, m)2^{l-1} \gcd(q-1, m)2^{l-1}}{(p-1)(q-1)}$$

so

$$\begin{aligned}
& \frac{1}{4} < \frac{2 \gcd(p-1, m) 2^{l-1} \gcd(q-1, m) 2^{l-1}}{(p-1)(q-1)} \\
& \iff \frac{1}{2} < \frac{\gcd(p-1, m) 2^l \gcd(q-1, m) 2^l}{(p-1)(q-1)} \\
& \iff \frac{p-1}{\gcd(p-1, m) 2^l} \cdot \frac{q-1}{\gcd(q-1, m) 2^l} < 2.
\end{aligned}$$

Since the factors on the left hand side of this inequality are positive integers, they are both equal to 1. This implies that $p-1 = \gcd(p-1, m) 2^l$ and $q-1 = \gcd(q-1, m) 2^l$. It follows that the exact power of 2 dividing $p-1$ as well as the exact power of 2 dividing $q-1$ are equal to 2^l and the odd parts of $p-1$ and $q-1$ divide m . Denote the odd parts of p and q as $\tilde{p} := \gcd(p-1, m)$ and $\tilde{q} := \gcd(q-1, m)$ respectively. Then $p = \tilde{p} 2^l + 1$ and $q = \tilde{q} 2^l + 1$. Now $2^k m = n-1 = pq-1 = p(q-1) + p-1 = p(\tilde{q} 2^l) + \tilde{p} 2^l = 2^l(p\tilde{q} + \tilde{p})$. It follows $2^{k-l} m = p\tilde{q} + \tilde{p}$. Since $\tilde{q} | m$ we have that m is a multiple of $\tilde{q}$, namely $m = b\tilde{q}$ for some positive integer b . Therefore, we have $\tilde{p} = 2^{k-l} b\tilde{q} - p\tilde{q} \iff \tilde{p} = \tilde{q}(2^{k-l} b - p)$. Given that $k \geq l$, we have that $2^{k-l} b - p$ is an integer and $\tilde{q}$ divides $\tilde{p}$. By symmetry, $\tilde{p}$ divides $\tilde{q}$, which implies $\tilde{p} = \tilde{q}$ and contradicts the fact that $p \neq q$. Therefore, $t = 1$ and $n = p^a$ for some prime $p \geq 3$ and $a \geq 2$. The inequality in (1) now says that $p^{a-1} \leq 4$, so that $p = 3$ and $a = 2$, implying $n = 9$, and contradicts the initial hypothesis that $n > 9$.

We conclude that $\frac{|B|}{\varphi(n)} \leq \frac{1}{4}$. □

In hindsight, the Miller-Rabin test offers a more reliable evaluation compared to Fermat's test, yielding a more precise bound in probabilistic primality testing.

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